

Primary VPT Chain: ProducerUA -> ProZorro -> Retail

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

standardized_type=sour_cream

retailer=silpo_novus

promo_variant=promo_controlled

frequency=daily

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo_novus. Combined rule: daily median of available silpo and novus standardized_type prices.

Table: PreTests

series	adf_level_p	kpss_level_p	adf_diff1_p	kpss_diff1_p	adf_diff2_p	kpss_diff2_p	integration_class	stability_flag	standardized_type	retailer	promo_variant	cointegration_p	pair_n_obs
producer	0.014081535169907735	0.01	4.706435460154828e-13	0.1	3.085032729165042e-21	0.1	I(1)	1.0	sour_cream	silpo_novus	promo_controlled	nan	nan
prozorro	0.21880956753864628	0.04947047925942609	2.411890357322739e-12	0.1	3.969186107838014e-13	0.1	I(1)	0.0	sour_cream	silpo_novus	promo_controlled	nan	nan
retail	0.004287500858491388	0.1	1.239928896263484e-09	0.1	0.019263927527483436	0.1	I(0)	0.0	sour_cream	silpo_novus	promo_controlled	nan	nan
producer_to_prozorro	nan	nan	nan	nan	nan	nan	pair	nan	sour_cream	silpo_novus	promo_controlled	0.3418935830778411	178.0
prozorro_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	sour_cream	silpo_novus	promo_controlled	1.3652144856597731e-05	44.0
producer_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	sour_cream	silpo_novus	promo_controlled	0.8616243548316272	52.0

Table: ModelCoefficients

standardized_type	retailer	promo_variant	frequency	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	notes	primary_chain_doc
sour_cream	silpo_novus	promo_controlled	daily	producer_to_prozorro	ARDL	prozorro	producer	178	0.1503321327484457	0.53201805897649	nan	nan	nan	nan	nan	ok	0	bounds_lower_p=0.1226; bo...	Primary chain is strictly...
sour_cream	silpo_novus	promo_controlled	daily	producer_to_prozorro	NARDL	prozorro	producer	176	0.8049124174362647	0.3181803694069162	-0.9103202976511395	1.2936168519519301e-25	0.2798213093974368	0.3401073440796139	nan	unreliable	1	Wald short/long asymmetry...	Primary chain is strictly...
sour_cream	silpo_novus	promo_controlled	daily	prozorro_to_retail	ARDL	retail	prozorro	44	-9.601011836171608	0.07694171411926477	nan	nan	nan	nan	nan	ok	0	bounds_lower_p=nan; bound...	Primary chain is strictly...
sour_cream	silpo_novus	promo_controlled	daily	prozorro_to_retail	NARDL	retail	prozorro	42	-0.1980200475167872	0.14112645504642918	-0.8511038403320891	0.0001333165872720088	0.3403351008029207	0.2937145039594696	nan	ok	0	Wald short/long asymmetry...	Primary chain is strictly...

Table: ResidualDiagnostics

model_family	link	y_series	x_series	ljungbox_p	arch_p	jb_p	unreliable_flag
ARDL	producer_to_prozorro	prozorro	producer	0.9991919902903724	0.4332665662567991	0.028152037118155036	0
NARDL	producer_to_prozorro	prozorro	producer	0.0725257366713705	0.012783066753899433	0.03261969684847455	1
ARDL	prozorro_to_retail	retail	prozorro	0.9964814814282824	1.0	3.1005165099028315e-18	0
NARDL	prozorro_to_retail	retail	prozorro	0.9930559522004219	0.9939506568368194	1.1001876853366829e-14	0

Table: SeriesUsed

date	producer	prozorro	retail	standardized_type	retailer	promo_variant	combined_rule
2021-02-01	12.7168318078645	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-02	12.72632554286918	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-03	12.73569490777093	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-04	12.74493705667296	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-05	12.75404914367847	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-06	12.76302832289066	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-07	12.77187174841275	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-08	12.78057657434793	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-09	12.78861583935331	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-10	12.79582084628455	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-11	12.80272614554264	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-12	12.80986628752861	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-13	12.81777582264346	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-14	12.8269893012882	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-15	12.83804127386385	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-16	12.85602638141248	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-17	12.88286036370906	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-18	12.91430941925396	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-19	12.94613974654752	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-20	12.97411754409009	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-21	12.99400901038205	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-22	13.00158034392373	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-23	12.85209968138777	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-24	12.46659692011084	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-25	11.93948089958932	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-26	11.3651604593196	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-27	10.83804443879808	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-02-28	10.45254167752114	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-01	10.30306101498519	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-02	10.3211976511164	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-03	10.367971081139	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-04	10.43192658749644	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-05	10.50160945263215	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-06	10.56556495898959	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-07	10.61233838901219	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-08	10.6304750251434	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-09	10.62692996199335	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-10	10.6172401963284	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-11	10.60282386382634	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...
2021-03-12	10.58509910016495	nan	nan	sour_cream	silpo_novus	promo_controlled	daily_median_of_available...

Table: LagProfile

	pair	lag	corr
producer_to_prozorro	1	0.07433869937437788	
producer_to_prozorro	2	0.06992193974139833	
producer_to_prozorro	3	0.05247780353970976	
producer_to_prozorro	4	0.03707093523380621	
producer_to_prozorro	5	0.04488476174629457	
producer_to_prozorro	6	0.03942159255359992	
producer_to_prozorro	7	0.04774350069192749	
producer_to_prozorro	8	0.055533922878419725	
producer_to_prozorro	9	0.05815494362517447	
producer_to_prozorro	10	0.05835686030753	
producer_to_prozorro	11	0.05572808262664563	
producer_to_prozorro	12	0.05132684090763728	
producer_to_prozorro	13	0.0478799443885565	
producer_to_prozorro	14	0.0459078796867533	
producer_to_prozorro	15	0.04627227965570159	
producer_to_prozorro	16	0.04980743752852967	
producer_to_prozorro	17	0.05457381656925348	
producer_to_prozorro	18	0.05403494026121064	
producer_to_prozorro	19	0.045342075346488105	
producer_to_prozorro	20	0.030030963601459588	
producer_to_prozorro	21	0.011140301716297355	
producer_to_prozorro	22	-0.006115614807159968	
producer_to_prozorro	23	-0.017626157690186797	
producer_to_prozorro	24	-0.022831222527256615	
producer_to_prozorro	25	-0.024681880076761423	
producer_to_prozorro	26	-0.028172449275670572	
producer_to_prozorro	27	-0.03484368926283701	
producer_to_prozorro	28	-0.04221286356527852	
producer_to_prozorro	29	-0.04827032698871696	
producer_to_prozorro	30	-0.05291303035138733	
prozorro_to_retail	1	-0.061941865389452976	
prozorro_to_retail	2	-0.02667578465422984	
prozorro_to_retail	3	0.043564940710425645	
prozorro_to_retail	4	0.057208793883784956	
prozorro_to_retail	5	-0.04675250238063316	
prozorro_to_retail	6	-0.062461473280118	
prozorro_to_retail	7	0.022730244105623125	
prozorro_to_retail	8	0.17441464979084656	
prozorro_to_retail	9	0.1516195523911158	
prozorro_to_retail	10	0.18300370120517467	

Table: ModelEligibility

standardized_type	retailer	promo_variant	frequency	integration_producer	integration_prozorro	integration_retail	n_obs_producer	n_obs_prozorro	n_obs_retail	n_obs_pair_prod_prozorro	n_obs_pair_prozorro_retail	any_i2	eligibility_note
sour_cream	silpo_novus	promo_controlled	daily	I(1)	I(1)	I(0)	1793	188	54	178	44	0	Admissible for ARDL/ECM s...

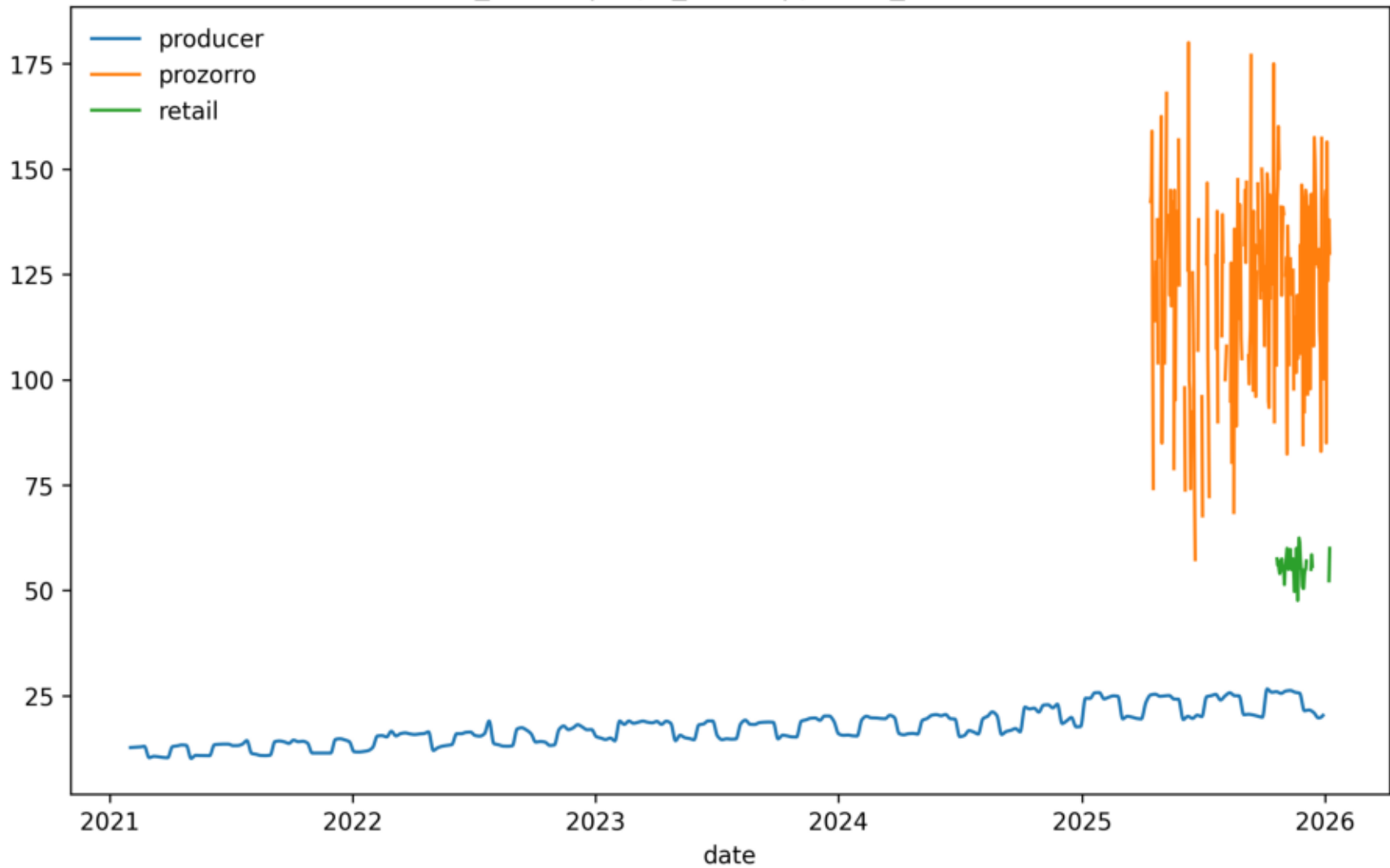
Table: NARDL_Multipliers

horizon	mult_pos		mult_neg	link
0	0.5272629908128263	-0.27764942662343833	producer_to_prozorro	
1	0.5660151486488385	-0.29805577903403124	producer_to_prozorro	
2	0.5688633093639157	-0.2995555812262039	producer_to_prozorro	
3	0.5690726401447521	-0.2996658119313484	producer_to_prozorro	
4	0.5690880252933407	-0.29967391353862705	producer_to_prozorro	
5	0.5690891560529745	-0.29967450898107123	producer_to_prozorro	
6	0.5690892391602219	-0.2996745527442027	producer_to_prozorro	
7	0.569089245268341	-0.29967455596065407	producer_to_prozorro	
8	0.5690892457172685	-0.2996745561970531	producer_to_prozorro	
9	0.5690892457502632	-0.2996745562144276	producer_to_prozorro	
10	0.5690892457526882	-0.2996745562157046	producer_to_prozorro	
11	0.5690892457528665	-0.29967455621579847	producer_to_prozorro	
12	0.5690892457528796	-0.29967455621580535	producer_to_prozorro	
13	0.5690892457528804	-0.29967455621580585	producer_to_prozorro	
14	0.5690892457528806	-0.2996745562158059	producer_to_prozorro	
15	0.5690892457528806	-0.2996745562158059	producer_to_prozorro	
16	0.5690892457528806	-0.2996745562158059	producer_to_prozorro	
17	0.5690892457528806	-0.2996745562158059	producer_to_prozorro	
18	0.5690892457528806	-0.2996745562158059	producer_to_prozorro	
19	0.5690892457528806	-0.2996745562158059	producer_to_prozorro	
20	0.5690892457528806	-0.2996745562158059	producer_to_prozorro	
0	0.03211818494801855	0.23013823246480575	prozorro_to_retail	
1	0.030386180358006807	0.2177278028091031	prozorro_to_retail	
2	0.03047958040239527	0.21839704738697954	prozorro_to_retail	
3	0.03047454371349613	0.21836095771744815	prozorro_to_retail	
4	0.03047481532187954	0.21836290388823523	prozorro_to_retail	
5	0.030474800675131367	0.21836279893906885	prozorro_to_retail	
6	0.030474801464971613	0.21836280459855548	prozorro_to_retail	
7	0.0304748014223787	0.2183628042933621	prozorro_to_retail	
8	0.03047480142467557	0.21836280430981997	prozorro_to_retail	
9	0.030474801424551706	0.21836280430893246	prozorro_to_retail	
10	0.030474801424558384	0.2183628043089803	prozorro_to_retail	
11	0.030474801424558023	0.21836280430897773	prozorro_to_retail	
12	0.030474801424558044	0.21836280430897786	prozorro_to_retail	
13	0.030474801424558044	0.21836280430897786	prozorro_to_retail	
14	0.030474801424558044	0.21836280430897786	prozorro_to_retail	
15	0.030474801424558044	0.21836280430897786	prozorro_to_retail	
16	0.030474801424558044	0.21836280430897786	prozorro_to_retail	
17	0.030474801424558044	0.21836280430897786	prozorro_to_retail	
18	0.030474801424558044	0.21836280430897786	prozorro_to_retail	

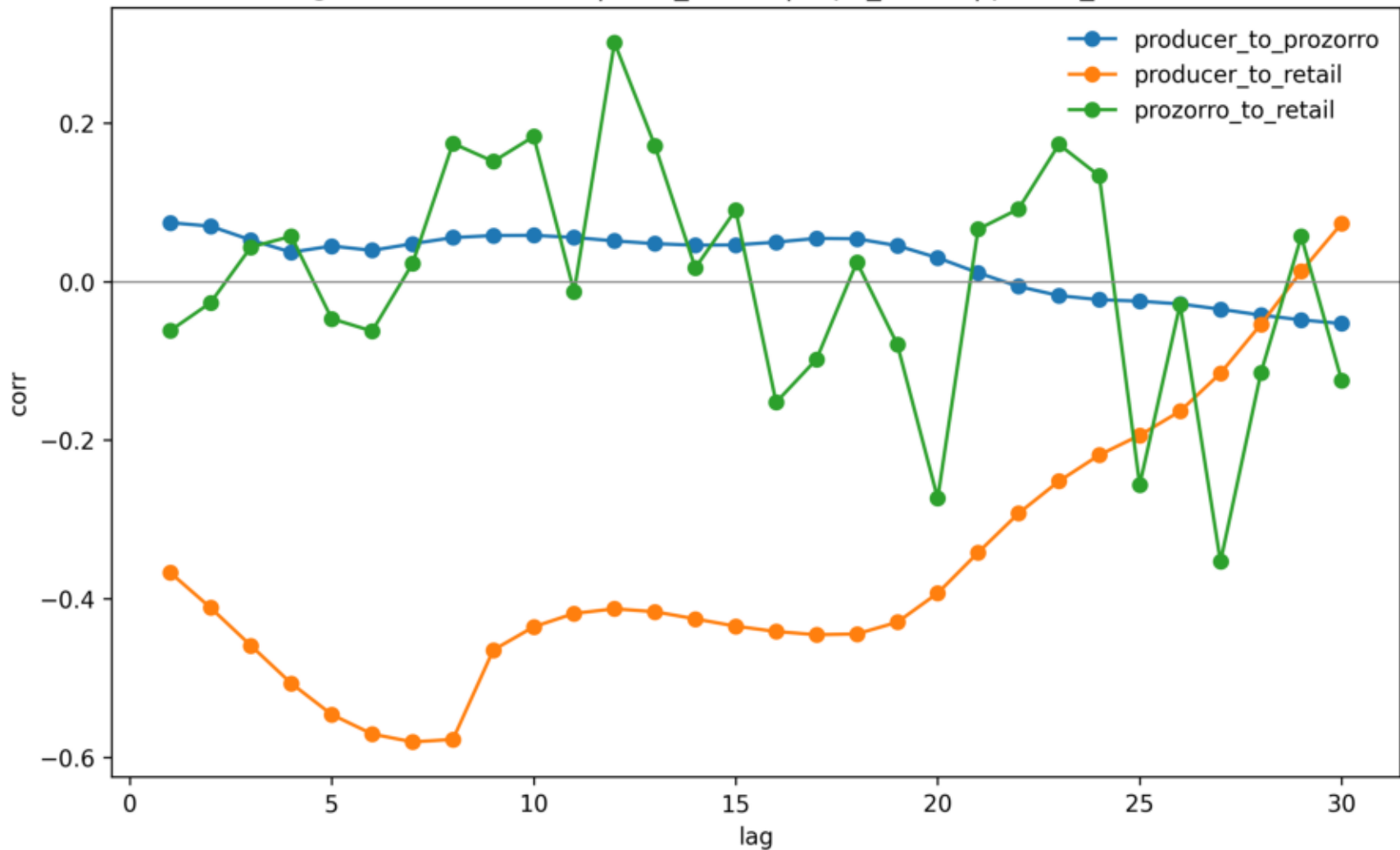
Table: VECM_IRF

No rows

sour_cream | silpo_novus | promo_controlled



Lag-Correlation Profile | sour_cream | silpo_novus | promo_controlled



NARDL Dynamic Multipliers | sour_cream | silpo_novus | promo_controlled

